Student-Faculty Co-Creation of Open Educational Resources for Learning Applied Statistics with Open Source Software Tools

[1]Nurlana Alili and
[2]Bryan Xu

[1,2] Department of Mathematical and Computational Sciences, University of Toronto Mississauga,
3359 Mississauga Road, Mississauga ON, L5M 3Z3

e-mail: [1] nurlana.alili@mail.utoronto.ca
[2] bryansuxi.su@mail.utoronto.ca

Abstract -

In most university courses, students learn from textbooks they did not help develop. We present an innovative approach where undergraduate students collaborated with faculty to create an open-access, interactive web-based e-book for an introductory applied statistics course. This process transforms undergraduates into co-authors rather than passive readers, fosters an appreciation for reproducible research, encourages academic collaboration, and offers hands-on opportunities to acquire and apply new skills and technologies. Faculty also benefit from fresh ideas and new perspectives. This type of collaboration is important to explore since evidence on student-faculty partnerships in developing course materials is sparse in the literature, particularly in a Canadian context and in the area of statistics. In this paper, we illustrate the contributions of undergraduate students to curricular innovation in the development of the e-book by describing the development process, methodology, and software tools used, and reflect on our experience as collaborators alongside faculty. Our participation deepened our learning while producing resources to benefit future learners. We highlight the potential of open-source technologies and student–faculty collaboration to support the development of inclusive, adaptive, and accessible resources for statistical education.

***Keywords*:** Bookdown, statistics education, curriculum development, interactive technology

1 INTRODUCTION

Undergraduate courses designed through a partnership between faculty and students can deepen engagement and develop a meaningful learning experience [2], [4]. Students are provided with the opportunity to be co-creators, rather than simply passive recipients of information, thereby encouraging them to take ownership of their educational journey, and faculty gain fresh insights into how students learn [4]. These partnerships deepen engagement and promote more authentic learning experiences for both students and faculty involved [1], [4]. Systematic reviews highlight that such collaborations enhance motivation, empower students, and enrich teaching practice; however, they have not been studied or documented uniformly across disciplines [2], [6]. Reports of whole-course co-creation are particularly limited within statistics education [5]. Systematic reviews by Mercer et al. [6] and Omland et al. [7] discuss several studies where students and faculty have partnered in course design internationally. However, there is a lack of such collaboration documents about mathematics and statistics education [3]. To the best of our knowledge, we could not find any collaborations specifically in a Canadian context.

Furthermore, many undergraduate statistics courses introduce foundational concepts through traditional lectures and texts. Even digital versions of textbooks are still typically static, offering limited advantages beyond the ability to quickly search for content and conveniently apply bookmarks. While these approaches build necessary theoretical understanding, they are often unable to provide opportunities for students to interact meaningfully with statistical methods or explore real-world applications. Topics such as hypothesis testing, confidence intervals, and regression may frequently remain abstract due to limited context and passive delivery formats [9].

Our project emerged in response to the lack of study in student-faculty collaboration in co-developing course material and to simultaneously address the limitations of traditional texts in statistics. We partnered with faculty through the *STA378: Independent Research Project Course*, at the University of Toronto Mississauga (UTM), to assist the current course coordinator, for *STA258: Statistics with Applied Probability*, develop an interactive e-book for the course as part of an ongoing course redevelopment project. STA258 is an introductory applied statistics course at UTM where most students learn about applied inference techniques (i.e., testing of hypotheses, interval estimation) and regression for the first time in their university learning experience. This is an important course since it provides students with the foundational skills for data analysis and equips them with the ability to interpret statistical results in advanced courses across all disciplines, including other statistics courses, the life sciences, social sciences, economics, finance, and computer science courses requiring a knowledge of inference. This course also provides students with the ability to read and interpret statistical results from academic journals, which is essential for critically evaluating research, making evidence-based decisions, and succeeding in advanced study and professional practice in both academic and non-academic settings.

The current course coordinator is involved in a long-term project to redesign the content in STA258. There are several reasons justifying the development of STA258. One such reason is that it currently overlaps considerably with *STA260: Probability and Statistics II*, with both courses sharing common introductory content. Students simultaneously enrolled in both courses experience substantial duplication in the content delivered to them during the opening weeks of both courses. Another reason to redevelop STA258 is the necessity to incorporate more data analysis components to modernize the course and make it more relevant for students by equipping them with foundational skills for independent statistical analysis. There are some distinct features of STA258 that make it a good candidate for collaborative re-design between students and faculty. Within the University of Toronto system, STA258 is only offered at the Mississauga campus, and there is no equivalent course or combination of courses available at the St. George campus or the Scarborough campus. Therefore, it has the potential to be a self-contained course that can equip students with the skills required, providing an efficient pathway for students to learn important skills related to statistical reasoning, which is equivalent to multiple courses from other campuses. Redesigning STA258 eliminates the necessity for the Department of Mathematical and Computational Sciences at UTM to go through the lengthy development and approval process required for an entirely new course to address the curricular gap.

The STA256 course is an introduction to probability and statistics, which provides the basic understanding necessary for STA258. In STA256, you will work with probability distributions, expectation, and the behavior of discrete and continuous random variables and vectors. While you'll be introduced to distribution functions, transformations of random variables, limit theorems, and the central limit theorem, the course is designed to give you the basic theory and language of probability theory, necessary to apply and extend what you use in STA258 statistical methodologies. STA260 is not a requirement for enrollment in STA258, but it does overlap in content. STA260, which emphasizes theoretical approaches and random sampling methods based on its predecessors, recognizes that STA258 draws upon the theoretical foundations and sampling distributions introduced in STA256. Other topics like point and interval estimation, hypothesis testing, and their relationships are also treated in STA258. However, STA260 delves further into the theoretical underpinnings than STA258, allied tests, unbiasedness, consistency, and sufficiency. The overlap between STA258 and STA260 emphasizes that one can learn statistics in different but cohesive ways across the curriculum, not just in an applied, statistical methodology approach in STA258, but also in making sense of broader theories in STA260 and formal, proof-based approaches for understanding statistical methodologies.

The core team working on this project consisted of the two undergraduate authors of this paper and two faculty members who are both Assistant Professors in Statistics at the Department of Mathematical and Computational Sciences at UTM. One of the faculty members is the current course coordinator for STA258. We contributed to content development and the technical implementation, and the faculty members provided consistent guidance through regular meetings, detailed feedback, training with software, and contributed critical input into decisions regarding the e-book's structure and tools. Both Professors also assisted with troubleshooting, and the course coordinator contributed a substantial amount towards the coding component, particularly with initiating the framework and resolving some of the more challenging components. We also received valuable feedback from two additional external partners who provided suggestions related to content scope, pedagogical structure, and overall educational value. One of the external partners is faculty at a College in British Columbia, and the other is an expert consultant in communication and design of digital assets. We met with faculty weekly for progress checks and troubleshooting, and with our external partners approximately once a month to provide progress updates and to receive feedback. One of the authors was living outside of Canada and joined meetings through video conference.

During the initial phase of the STA258 course development timeline, we as undergraduate students worked alongside Dr. Mudalige and Dr. Ataei to produce an open-access, interactive e-book which integrates narrative exposition, reproducible simulations, and examples involving code, embeds visualizations and interactive components to explain concepts, and provides an open-source resource for educators in the broader statistical community. This paper outlines the framework that guided our partnership, describes the authoring workflow (version control, peer review, and continuous deployment), and reflects on early outcomes. By sharing these first-phase results, we aim to provide a practical model for implementing student-faculty co-creation in applied statistics and related quantitative courses.

2 DESIGN, DEVELOPMENT AND IMPLEMENTATION

The interactive e-book project was launched during the Summer 2025 semester (May-August) as a joint initiative between undergraduate students and faculty to create an engaging and innovative learning resource for students enrolled in statistics courses. The University of Toronto Mississauga provides high-performing undergraduate students enrolled in a statistics major or specialist with the opportunity to enrol in *STA378: Statistics Research Project*, where students explore a topic in statistics under the supervision of a faculty member. Students can not directly enrol in STA378 through ACORN, and a special administrative process is required, including approval from the associate chair. Both authors of this paper had recently completed STA258 during the Winter 2025 semester and agreed with Dr. Mudalige that the course required adjustments. We mutually expressed interest in working together on this course redevelopment project.

Our initial decision was to use LaTeX to create a Portable Document Format (PDF) e-book. This was based on its well-established typesetting capabilities and familiarity within academic publishing. This PDF document, generated in LaTeX, is also easily accessible on many devices. In addition, open-source resources, such as OpenIntro Statistics, offer extensive material to utilize as a foundation to develop an e-book using LaTeX. However, there are limitations with this approach, including being limited to fixed-sized pages to display content, limited navigation options, difficulty with embedding interactive components, and maintaining an organized working environment due to the many auxiliary files produced.

Although we began the e-book in LaTeX, midway through our development, we learned about Bookdown, which is an R package developed for writing materials such as books and long-form content with R Markdown. Bookdown has the advantage of converting R Markdown into responsive HTML, EPUB, and print-ready PDF while automatically generating cross-references and a searchable table of contents, LaTeX can be typeset within R Markdown documents, the HTML output from Bookdown can embed live R code, interactive widgets, and interactive apps using libraries such as shiny and plotly, giving students a hands-on e-book experience that fixed-layout PDFs cannot match, and a book generated in Bookdown can be hosted on GitHub pages allowing anyone with an internet connection to easily access it. LaTeX code written in Bookdown is rendered using MathJax, which is an open-source JavaScript library for mathematical symbols and equations for web browsers. Therefore, the vast majority of content that can be typeset in LaTeX can be rendered on a browser using Bookdown. Due to its many advantages over traditional LaTeX, we decided to transition to the Bookdown framework since it proved more compatible with the goals of interactive and accessible online learning.

This shift from LaTeX to Bookdown allowed us to integrate dynamic and interactive elements that enhance student engagement and understanding, and transformed static content into a fully interactive learning resource. The conversion from LaTeX (.tex) to R Markdown (Rmd) was performed using Pandoc, with the majority of the formatting and structure intact. Pandoc also preserved most of the fidelity of the LaTeX code during the conversion. The syntax used for the conversion is given in Listings 1 and 2.

```
pandoc -s input.tex -f latex -t markdown -o output.Rmd
```

Listing 1: Basic conversion of a LaTeX .tex file to an R Markdown .Rmd file using Pandoc.

```
pandoc --filter pandoc-citeproc --mathjax -s input.tex -f latex -t markdown -o
    output.Rmd
```

Listing 2: Conversion of a LaTeX .tex file to an R Markdown .Rmd file rendered with MathJax using Pandoc.

Once we established our framework, we selected other open-source tools and programming languages that could support both visual design and reproducible workflows around Bookdown. A summary of the technologies used in developing the e-book is listed in Table 1.

Languages	R, Markdown, LaTeX, YAML, HTML, CSS, JavaScript
Frameworks	Bookdown, R Markdown
R packages	`knitr`, `ggplot2`, `plotly`, `Shiny`, `rgl`, `dplyr`
Version Control	Git (with GitHub)
Deployment	GitHub Pages, Shinyapps.io

Table 1: Summary of software, libraries and frameworks used in developing the e-book.

The tools utilized in Table 1 were intentionally selected for their compatibility, transparency, and capacity to support reproducible computing. Some tools were used directly, such as R Markdown and R, while others were secondary technologies that were used by the frameworks in rendering the e-book. GitHub served both as a collaborative development platform and as a public-facing repository for version control, enabling open access to the resources created and providing asynchronous contributions from others. The core members worked on developing content on various branches on GitHub before merging everything onto the main branch after sufficient progress. We utilized GitHub Pages to host the e-book directly from the repository used for the project. GitHub Pages is a service offered by GitHub that allows users to host webpages directly from a GitHub repository. The YAML file was used for front matter, to store the book's metadata, for build settings, and integration with GitHub Pages. CSS was used to customize environments in Bookdown, such as the definition environment and remark environment, with personalized aesthetics and a consistent colour scheme.

Although we encountered some technical challenges, especially during the transition from LaTeX to Bookdown, these obstacles deepened our understanding of collaborative digital authorship and reproducible research. The final product represents not only an innovative approach to content delivery but also an exploration of what collaborative educational development can look like in practice.

The development of the e-book required significant planning and coordination to ensure that the final product was cohesive and pedagogically meaningful. The initial draft was largely a collection of lecture notes and presentation slides. We used these to build a single unified document, which was a one-to-one replication of the notes and slides. This was our baseline draft for the e-book. During the development of this baseline, the inclusion of figures and graphics was sparse since we were aware that content would be rearranged. After reviewing this initial baseline draft, we realized that the large number of chapters resulted in the fragmentation of related content, making it difficult for students to learn the material.

Through collaboration and discussion, we combined overlapping sections and reordered material into a more logical flow. This process helped reduce redundancy and resulted in a smoother experience for both students and instructors, and the material was presented coherently. We also removed as much content as possible that overlapped with STA260, making these two courses much more distinct. Once the order and layout were finalized, the content created was rearranged and refined, and static and interactive plots and figures were built around the updated content.

A brief description of the specific R packages from Table 1 which were used in developing the e-book is given in Table **??**:

R Package	**Description**
`knitr`	The knitr package is designed to integrate R code, output, and commentary into a single document, facilitating reproducible research and dynamic report generation.
`ggplot2`	A data visualization package that implements a layered grammar of graphics, enabling the systematic creation of a diverse range of static plots.
`plotly`	This package provides tools for generating interactive, web-based plots and dashboards that can be easily shared and embedded.
`Shiny`	An R package for building interactive web applications directly from R, allowing users to create powerful data analysis tools and visualizations without extensive knowledge of web development languages.
`rgl`	A visualization package that creates interactive 3D plots and scenes, providing a robust platform for exploring spatial data. It leverages OpenGL for high-performance rendering and can export scenes to WebGL, allowing for the display of 3D content within web browsers.

Table 2: Description of each R package used in developing the e-book.

Static and interactive plots and figures are rendered natively within the R Markdown file within code blocks. The Tikz package was initially used with LaTeX to create some static graphics, which were used in both the initial PDF and the first draft of the Bookdown version of the e-book, since Tikz is a powerful package that can create vector graphics. However, we discovered that we can create all of the required graphics using `ggplot2` and plotly, so we decided to utilize these 2 packages for all figures for a more consistent appearance. An example of a code block used to render a figure using the ggplot2 library is given in Listing **??**.

```
```{r fig_label, fig.cap="Example␣Code␣Block", echo=FALSE, fig.align='center', warning=FALSE}

library(ggplot2)
# Code for ggplot2 figure goes here
# The ggplot2 figure is natively created and embedded into HTML rendered
#  in Bookdown
```
```

Listing 3: An example of a code block in Markdown used to render an image with ggplot.

Interactive plots were created using Plotly and R Shiny. Plotly figures are generated natively within code blocks in a similar manner to ggplot2, with syntax similar to that of **??**. R Shiny apps were developed in R and hosted on Shinyapps.io, which is a platform provided for hosting and sharing Shiny web applications. Shinyapps.io is maintained by Posit, who are most well known as the developers of R Studio and Positron. These apps are natively embedded into the e-book using a code block and some HTML for formatting. Listing **??** provides the syntax for embedding R Shiny apps in the e-book.

```
```{r shiny_app_label, fig.cap="Example␣Interactive␣Application", echo=FALSE, out.width='100%', warning=FALSE}
```
```

````

knitr::include_app("URL to App Hosted on shinyapps.io")
```
````

Listing 4: An example of a code block in Markdown used to render an image with ggplot

Finally, we used the rgl library to create a 3D interactive version of the digital hex sticker designed for the project, which appears on the cover of the e-book. This was done for aesthetic purposes and to provide the e-book with a recognizable identity.

3 CONTENT AND INTERACTIVE DESIGN

The textbook is live and available to be accessed at https://nishanmudalige.github.io/STA258/. We note that the e-book is still a work in progress since it is one component that is part of an ongoing, long-term course redevelopment project, which we discussed in Section **??**.

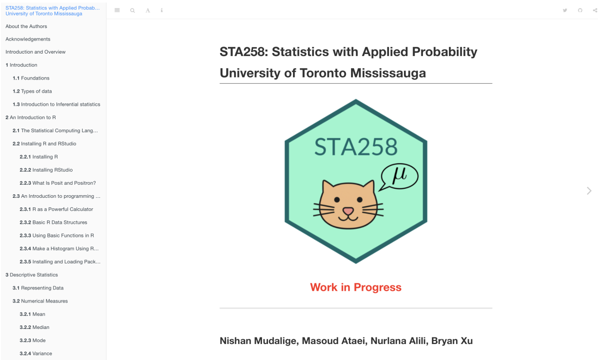
We selected the default theme of Bookdown since it has a clean layout and appearance. The default theme also comes with options for light mode, dark mode, and high contrast (sepia) mode, with the table of contents appearing along the left. The landing page in light and dark mode is presented in Figures 1a and 1b. The table of contents is available on the left of the webpage, and it can be hidden if desired using the menu button on the top menu bar.

There is also a search feature where indexing is automatically performed by Bookdown. The search feature is accessed from a magnifying glass icon near the top of the page. Additional accessibility features to help improve legibility include the ability to increase or decrease font size, increase or decrease spacing between lines, and the ability to switch between serif and sans serif fonts. The search feature and accessibility options are illustrated in Figures 1c and 1d

The main chapters in the e-book are organized in the following manner:

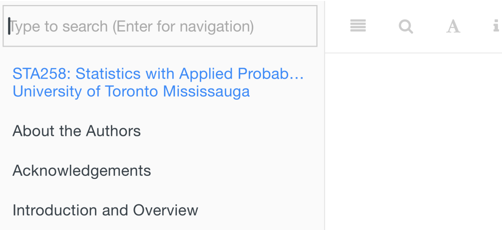
- Chapter 1: Introduction
- Chapter 2: An Introduction to R
- Chapter 3: Descriptive Statistics
- Chapter 4: Foundations of Inference
- Chapter 5: Confidence Intervals
- Chapter 6: Hypothesis Tests
- Chapter 7: Statistical Power
- Chapter 8: Analysis of Variance
- Chapter 9: Simple Linear Regression
- Chapter 10: Analysis of Categorical Data

Chapter 1 is a broad introduction to applied statistics. Chapter 2 is a dedicated section with an introduction to R. Chapter 3 contains foundational statistical concepts, including key terminology, data types, and the basic ideas behind inferential statistics. Chapter 4 sets the stage for statistical inference, covering sampling distributions and the main reference distributions in the course (the normal distribution, t-distribution, F-distribution, and chi-square distribution. Chapters 5 and 6 cover standard one-sample and two-sample inference techniques on a mean, on a proportion, on a difference of means, on a difference of proportions, on paired data, on a variance, and on a ratio of two variances. Chapter 7 discusses the type I and type II errors and introduces statistical power for the one-sample case when the population variance is known. Chapter 8 covers one-way analysis of
````

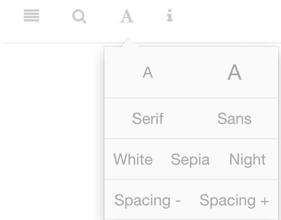


(a) Landing page of the e-book in the default layout.

(b) Landing page of the e-book in dark mode.



(c) Activating the search feature from the menu bar.

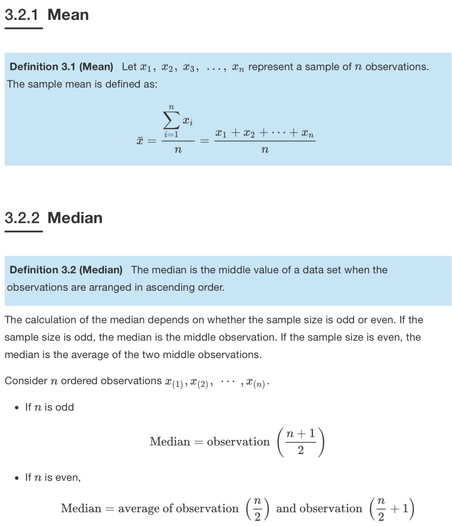


(d) Accessibility options accessed from the menu bar.

Figure 1: Landing page of the e-book in the default light theme and dark theme. Notice the table of contents in the left panel. Icons for changing theme, searching, changing font size, and spacing are located near the top left and are elaborated in the sub-figures.

variance. Chapter 9 covers simple linear regression, including inference on the slope. and Chapter 10 covers chi-square tests for categorical data. These chapters were further broken down at a more granular level into sub-section; however, the titles of these components are not listed for brevity.

Examples of the appearance of standard text-based content and mathematical statements which appear in the body of the e-book are presented in Figure 2, highlighting the use of coloured blocks to help distinguish several recurring components, such as definitions, remarks, and theorems. Notice that mathematical statements and symbols that are rendered with MathJax appear similar to content that would appear in a document rendered using LaTeX.



3.2.1 Mean

Definition 3.1 (Mean) Let $x_1, x_2, x_3, \ldots, x_n$ represent a sample of n observations. The sample mean is defined as:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

3.2.2 Median

Definition 3.2 (Median) The median is the middle value of a data set when the observations are arranged in ascending order.

The calculation of the median depends on whether the sample size is odd or even. If the sample size is odd, the median is the middle observation. If the sample size is even, the median is the average of the two middle observations.

Consider n ordered observations $x_{(1)}, x_{(2)}, \cdots, x_{(n)}$.

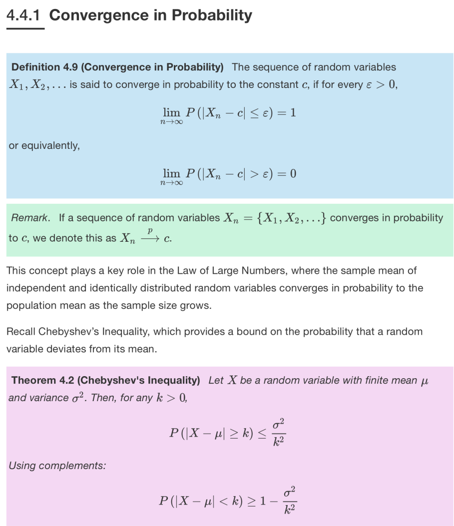
- If n is odd

$$\text{Median} = \text{observation} \left(\frac{n+1}{2}\right)$$

- If n is even,

$$\text{Median} = \text{average of observation} \left(\frac{n}{2}\right) \text{ and observation} \left(\frac{n}{2}+1\right)$$

(a) Sample content from the body of Chapter 3: Descriptive Statistics.



4.4.1 Convergence in Probability

Definition 4.9 (Convergence in Probability) The sequence of random variables $X_1, X_2, \ldots$ is said to converge in probability to the constant c, if for every $\varepsilon > 0$,

$$\lim_{n\to\infty} P\left(|X_n - c| \leq \varepsilon\right) = 1$$

or equivalently,

$$\lim_{n\to\infty} P\left(|X_n - c| > \varepsilon\right) = 0$$

Remark. If a sequence of random variables $X_n = \{X_1, X_2, \ldots\}$ converges in probability to c, we denote this as $X_n \xrightarrow{p} c$.

This concept plays a key role in the Law of Large Numbers, where the sample mean of independent and identically distributed random variables converges in probability to the population mean as the sample size grows.

Recall Chebyshev's Inequality, which provides a bound on the probability that a random variable deviates from its mean.

Theorem 4.2 (Chebyshev's Inequality) *Let X be a random variable with finite mean μ and variance σ^2. Then, for any $k > 0$,*

$$P\left(|X - \mu| \geq k\right) \leq \frac{\sigma^2}{k^2}$$

Using complements:

$$P\left(|X - \mu| < k\right) \geq 1 - \frac{\sigma^2}{k^2}$$

(b) Sample content from the body of Chapter 4: Foundations on inference.

Figure 2: Examples of content with text and mathematical statements in the e-book.

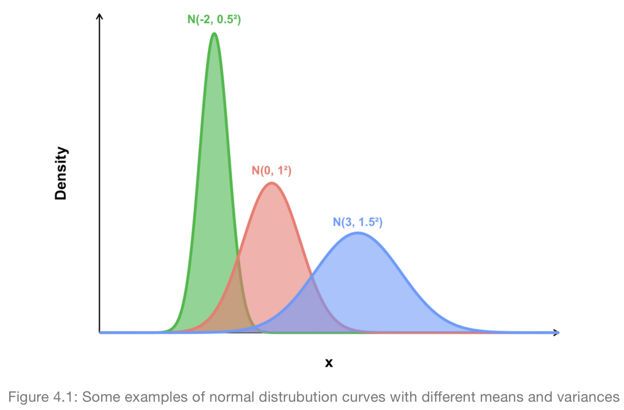
Recognizing the importance of technical skills, the second chapter is an introduction to R and its practical uses. This section explains how to install R, use essential functions, and work with basic data structures and visualizations. Together, these elements provide a strong technical foundation. Installation instructions for both R and R Studio are provided, including screenshots to help guide the reader. Information related to working with R Studio was included in this chapter since R Studio is a popular integrated development environment (IDE) for programming in R.

This chapter helps the reader become familiar with R by introducing the language as a powerful calculator and also introduces some basic data structures, functions, and graphical plots. This sets the necessary groundwork for the coding examples, which are provided in subsequent chapters.

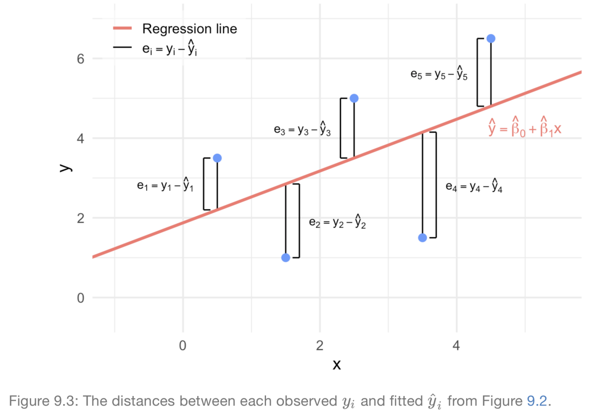
We also introduce Positron, which is the next-generation of data science IDE that also supports coding with Python. Positron was introduced in an attempt to potentially future-proof the e-book and to inform the reader that the creators of the e-book are aware of trends in programming with statistical software.

Numerous static figures were created using `ggplot2`, within code blocks in the form of Listing **??**. Some examples of the static figures created are given in Figure 3. We utilized the default colour palette of `ggplot2`, or emulated it where necessary for visual consistency, and also because this colour palette was developed with the principles of effective data visualization and accessibility in mind.

Although `plotly` can be used for interactive figures, we also used `plotly` to create several static figures by deactivating the options that allow for interactivity. This decision was made out of convenience for a more unified design of figures and aesthetics. Not all of the figures in the e-book required an interactive component. If interactivity were forcefully implemented on all figures, some of them would have become unnecessarily distracting for students. Examples of figures created with `plotly` in which the interactive options were removed are presented in Figure 4.

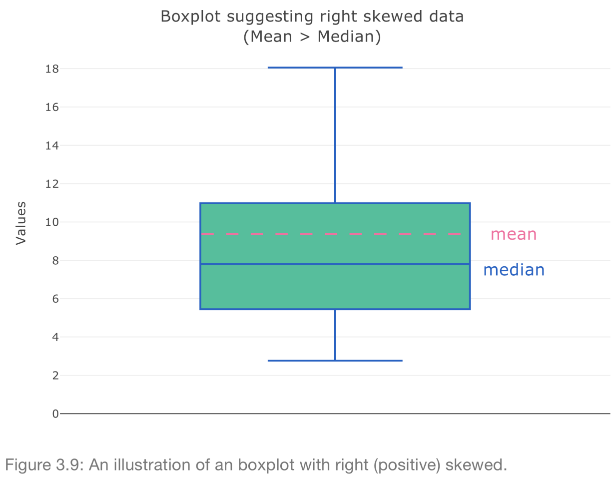


(a) Multiple normal distributions created with `ggplot2` from a figure in Chapter 4: Foundations on inference.

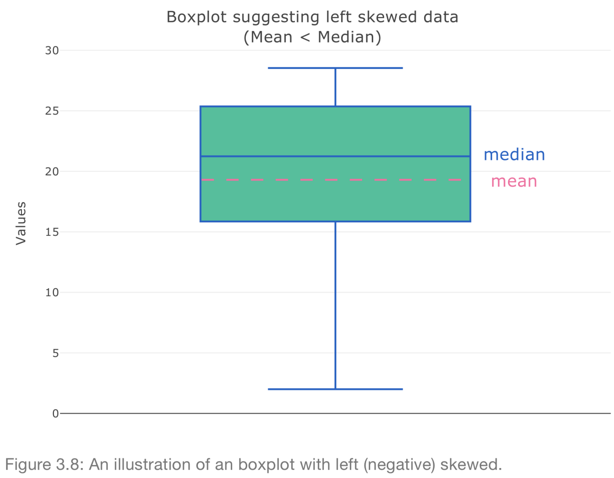


(b) A figure introducing residuals created using `ggplot2` from Chapter 9: Simple Linear Regression.

Figure 3: Examples plots created using `ggplot2`.



(a) A right-skewed boxplot created with `plotyle` with interactive elements disabled from Chapter 3: Descriptive Statistics.

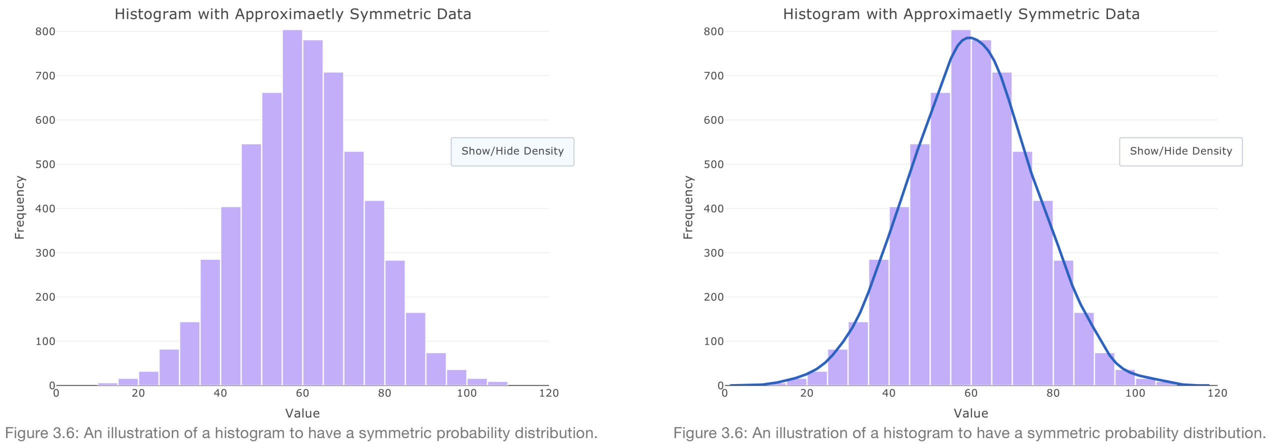


(b) A left-skewed boxplot created with `plotyle` with interactive elements disabled from Chapter 3: Descriptive Statistics.

Figure 4: Examples plots created with `plotly` where the interactive options were disabled.

One of the areas in which this e-book stands out is the inclusion of interactive plots. Incorporating interactive plots and interactive applications allows users to learn by applying themselves and observing dynamic changes. Some interactive plots were created using `plotly` while others were created using `shiny`. Plots with relatively simpler components were created with `plotly`, such as plots which would display or hide information when the user hovers over them with their cursor, and plots which had buttons on them to show or hide a feature. An example of an interactive plot created with `plotly` containing an interactive button is illustrated in Figure 5. An example of an interactive plot created with `plotly`, which dynamically changes based on the component of the plot where the user hovers with their cursor, is illustrated in Figure 6.

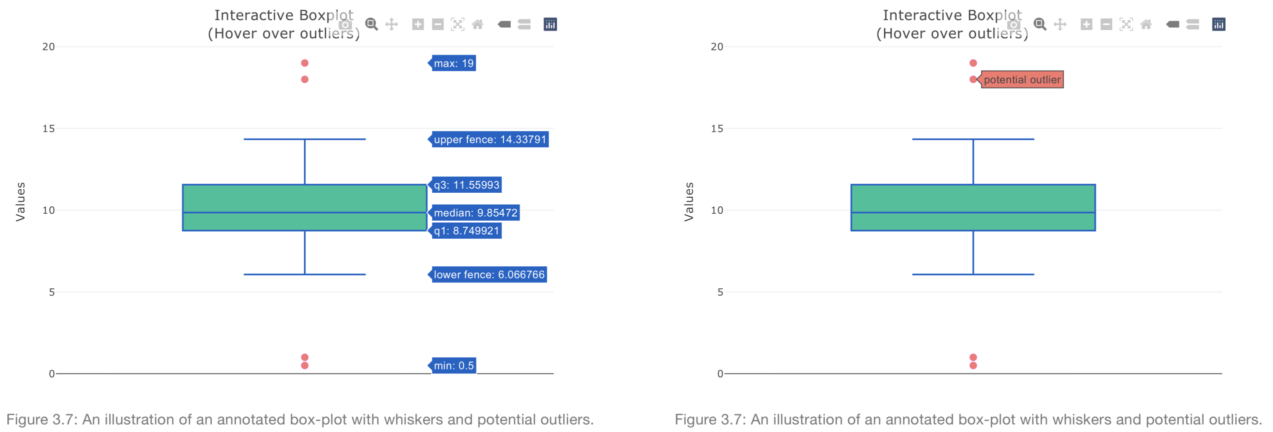
Applications that required more advanced interactive capabilities were implemented using `shiny`, with each shiny app being hosted on Shinyapps.io as discussed in Section **??**. The `shiny` package allows us to implement advanced features which are not possible in `plotly`, such as sliders, tabs, switches and control buttons,



(a) Interactive histogram created with `plotly` with density plot hidden from Section 3: Descriptive Statistics.

(b) Interactive histogram created with `plotly` with density plot shown from Section 3: Descriptive Statistics.

Figure 5: Interactive plot of a histogram created with `plotly`, which can show or hide the density plot superimposed over the histogram when the user clicks the *Show/Hide Density* button on the plot.

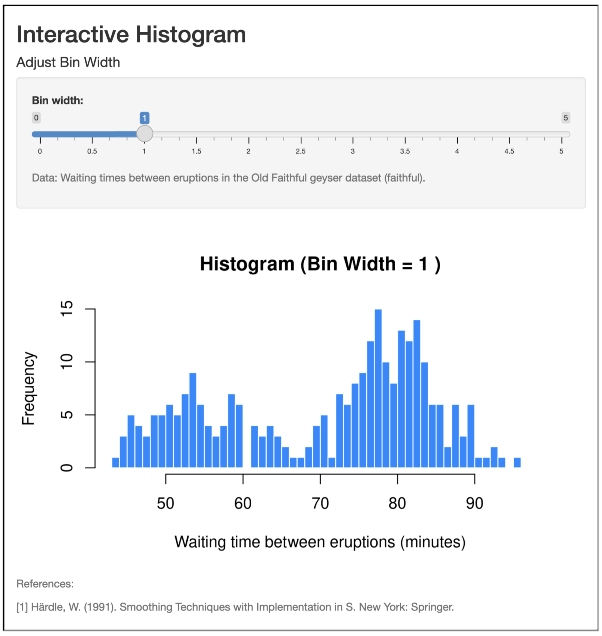


(a) Interactive boxplot created with `plotly` from Section 3: Descriptive Statistics. Information appears when the user moves the cursor over the plot.

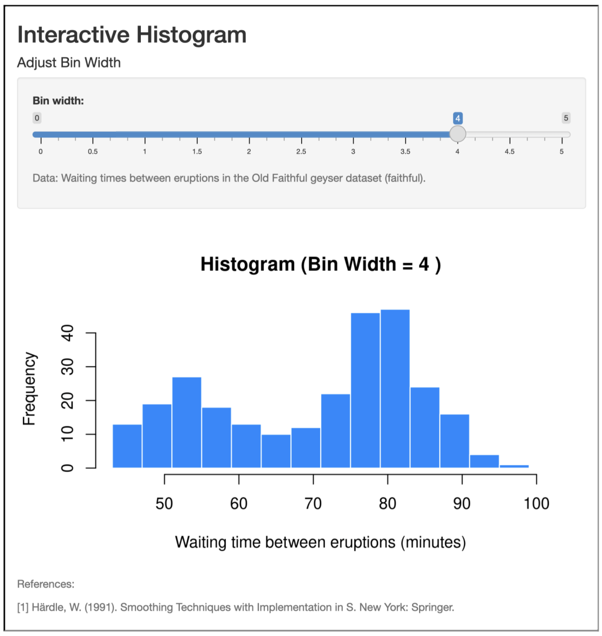
(b) Interactive boxplot created with `plotly` from Section 3: Descriptive Statistics. Information changes when a user identifies an outlier with their cursor.

Figure 6: Interactive plot of a boxplot created with `plotly` where the information changes based on the component of the plot where the user hovers over with their cursor. Note that in this particular example, the baseline plot does not display any text labels when the cursor does not hover over it.

menus, reading and writing files, check boxes, and input boxes for text and numbers. Although the time commitment required to refine a shiny app is considerably longer, the resulting product is very powerful. Some examples of interactive apps created using `shiny` are given in Figures 7 and 8.

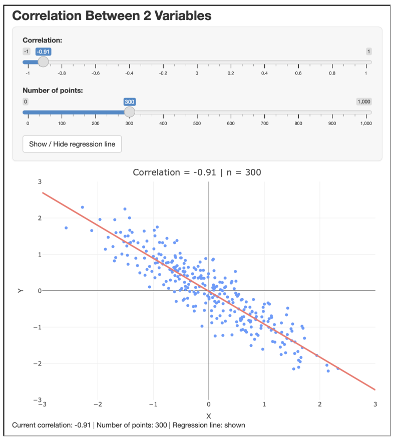


(a) Interactive histogram with the slider for bin width set to a bin width of 1 minute.

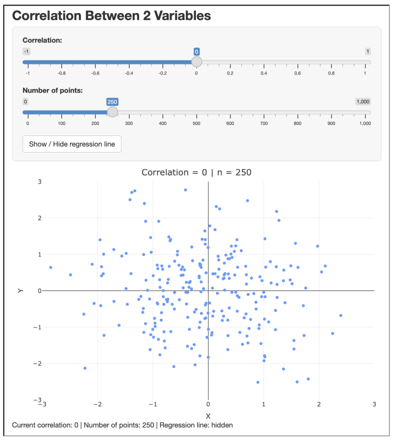


(b) Interactive histogram with the slider for bin width set to a bin width of 4 minutes.

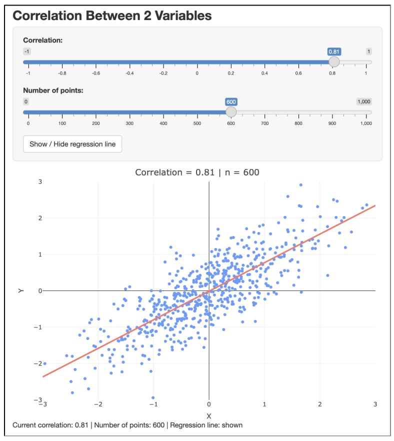
Figure 7: Interactive application created using `shiny` that allows students to adjust the bin width for a histogram, which was built with real-world data. This specific example used the *faithful* data set, which has waiting times for the Old Faithful geyser in Yellowstone National Park. The tool demonstrates how the appearance of a histogram is sensitive to graphical parameters, enhancing understanding of distribution shape and granularity. Notice the title of the histogram also updates based on the numeric value of the bin width the slider has been set to.



(a) Interactive plot created with `shiny` to demonstrate correlation between 2 variables, showing data with strong negative correlation with the regression line superimposed.



(b) Interactive plot created with `shiny` to demonstrate correlation between 2 variables showing a correlation of approximately 0.



(c) Interactive plot created with `shiny` to demonstrate correlation between 2 variables, showing data with strong positive correlation with the regression line superimposed.

Figure 8: Interactive application which simulates data points based on the correlation and the number of points selected. This allows the user to observe various strengths of positive and negative correlation. The user can also show or hide the ordinary least squares regression line on the plot.

At the end of each major topic in Chapters 3 – 10, we also included examples with complete solutions. Recall that Section 1 introduces the e-book, and Section 2 introduces R. Therefore, it was not necessary to include examples for these two chapters. Examples from Chapters 3 – 10 cover numerical calculations, interpretations, explanations, and many also include a coding component involving R. The examples can be studied by students to reinforce concepts covered in the book, to become familiar with mathematical expressions, formulas, terminology, and interpretation, and they also provide opportunities to practice coding with R.

4 CONCLUSIONS, REFLECTION AND FUTURE WORK

Through this project, undergraduate students partnered with faculty to develop an interactive e-book that covers the full content of a typical introductory applied statistics course, completing the work within a single semester using entirely open-source tools. The e-book is freely accessible to anyone with a web browser and an internet connection. Even after deciding to change the development framework midway through the project, we successfully leveraged tools to streamline the conversion process.

The completed e-book is available to students, and all source files and the framework used to create it are openly hosted on GitHub at https://github.com/nishanmudalige/STA258/, making them freely available to the broader statistical education community. Educators can easily fork the repository to create their version, adapting and personalizing the content to meet the specific needs of their institution's curriculum.

Beyond its technical and instructional outcomes, this project demonstrates the pedagogical potential of student-led authorship and design. By actively participating in the creation of educational content, students shift from consumers to contributors of knowledge, engaging more deeply with the subject matter. The process required not only statistical expertise and programming fluency, but also adaptability, design thinking, and critical reflection—skills applicable in both academic and professional contexts. Through collaborative authorship, the final product embodies negotiated decisions, iterative problem-solving, and responsiveness to peer and faculty feedback. More broadly, the project presents a replicable model of participatory curriculum development that integrates open-source tools, student initiative, and faculty mentorship to generate accessible, engaging, and adaptive learning resources.

The undergraduate students who participated also gained experience with a wide range of tools and skills that would not typically be covered in their formal coursework. We engaged in both independent and collaborative learning, developing the ability to work independently on technical tasks and also collaborating effectively to work on components of the project. In addition to learning both breadth and depth of software platforms and version-control systems, we were introduced to principles of project management, including setting milestones, tracking progress, coordinating deliverables, and troubleshooting. We also developed the ability to be creative with plots and graphics, applying principles of design to produce clear, engaging, and visually effective materials. Throughout the project, we benefited from close mentorship with faculty, receiving guidance on both technical and pedagogical aspects of the work. We also had the opportunity to interact with external partners, including meetings to gather feedback, discuss design considerations, and align the project with broader educational goals. This combination of technical, professional, and interpersonal skill development provided a learning experience that extended well beyond the boundaries of a traditional classroom. Technology also played a crucial role in enabling us to collaborate effectively across geographical and temporal boundaries. One of the authors participated remotely from a time zone many hours ahead of the project's primary base, which required careful scheduling, asynchronous communication, and disciplined time management. This arrangement allowed full productivity to be maintained and highlighted the flexibility of our collaborative tools in supporting consistent, high-quality contributions within a distributed, technology-mediated academic environment.

One of the most time-consuming components was designing figures and interactive plots. Although this was challenging, it was also one of the most rewarding aspects of this project. Many static plots were created using `ggplot2`, some static plots and many interactive plots were created using `plotly`, and all advanced applications were created with `shiny`. The figures and interactive components are now central features of the e-book, which help convey concepts and also distinguish our e-book from an e-book published in a document format, such as PDF or EPUB, or even the proprietary digital books offered by a publisher. They are visually clear, carefully styled, and interactive where appropriate. These visualizations reflect intentional design choices and help communicate complex ideas in a way that is both intuitive and accessible. Although this process was time-consuming, it was also one of the most enjoyable and educational parts of the project.

While the current version of the e-book successfully integrates core statistical ideas with interactive elements, there are several directions for future development. One important area to expand the content is to develop exercises at the end of each chapter for readers to practice, along with solutions. Some exercises should also include a computer programming component with data sets. Exercises can be added and updated over time by the course coordinator. Since the core content has been established, the inclusion of exercises can be performed in the next phase of the STA258 redevelopment timeline.

Improving accessibility and the overall user experience is another key goal. Possible next steps include optimizing the interface for mobile devices and developing visual themes with accessible colour palettes to target individuals with specific visual impairments, such as deuteranomaly (red-green colour vision deficiency) which is the most common forms of colour vision deficiency affecting approximately 3.66% of males (95% CI: 3.02 – 4.44) and 0.46% of females (95% CI: 0.30 – 0.69) worldwide, based on a recent systematic review [8].

From a technical perspective, future versions could explore more extensive use of shiny applications, where students would be able to interact with full datasets and generate real-time statistical output within the e-book. We would also like to explore ways in which we can utilize `rgl` to enhance content. In the long term, we hope to maintain and grow the project through continued collaboration between students and faculty. Building a clear versioning system, contributor documentation, and a guide for future authors would help ensure that the e-book remains a living resource that evolves with each new cohort.

ACKNOWLEDGMENTS

We would like to sincerely thank Professor Nishan Mudalige, who served as the principal supervisor for this project from the very beginning. His guidance, encouragement, and consistent support played a central role in shaping both the direction and success of the e-book. We are also grateful to Professor Masoud Ataei, whose input during the later stages of development helped refine both the technical and pedagogical aspects of the project. His feedback was thoughtful and valuable throughout.

We thank our external partners, Dr. Steve Tully, Mathematics Professor at Okanagan College, and Avanti Coonghe, Communications and Design Consultant at the United Nations (World Health Organization), for their guidance, input, and feedback on content and design. We are also grateful to the Department of Mathematical and Computational Sciences at the University of Toronto Mississauga for fostering an environment where student–faculty collaboration is encouraged, actively supported, and facilitated.

Finally, we acknowledge the open-source community. The tools, documentation, and shared knowledge developed and maintained by contributors around the world made the technical side of this project achievable and rewarding.

REFERENCES

[1] C. Bengtson, S. Frederiksen, B. Hansson, et al., "Working as partners: Course development by a student–teacher team," *Teaching and Learning Inquiry*, vol. 1, no. 1, pp. 8–24, 2013.

[2] C. Bovill, "Co-creation in learning and teaching: The case for a whole-class approach in higher education," *Higher Education*, vol. 79, no. 7, pp. 1023–1037, 2019.

[3] F. Duah and T. Croft, "Students as partners in mathematics course design," in *Proceedings of the CETL–MSOR Conference 2011*, 2011, pp. 45–52.

[4] M. Healey, A. Flint, and K. Harrington, "Students as partners: Reflections on a conceptual model," *Teaching and Learning Inquiry*, vol. 4, no. 2, pp. 8–20, 2016.

[5] C. Kim, J. Smith, and P. Lee, "Involving students as partners in a course redesign: A case study in statistics education," *Studies in Higher Education*, vol. 47, no. 4, pp. 567–582, 2022.

[6] L. Mercer-Mapstone, C. Bovill, P. Felten, et al., "A systematic literature review of students as partners in higher education," *International Journal for Students as Partners*, vol. 1, no. 1, pp. 1–18, 2017.

[7] M. Omland, L. Nygaard, I. Edvardsson, et al., "Co-creation in higher education: A conceptual systematic review," *Higher Education*, 2025. DOI: `10.1007/s10734-024-01364-1`.

[8] Y. D. Jeong et al., "Global prevalence of congenital color vision deficiency among children and adolescents, 1932–2022," *Ophthalmology*, 2025. DOI: `10.1016/j.ophtha.2025.07.031`.

[9] N. Tintle, B. Chance, G. Cobb, S. Roy, T. Swanson, and J. VanderStoep, "Combating anti-statistical thinking using simulation-based methods throughout the undergraduate curriculum," *The American Statistician*, vol. 69, no. 4, pp. 362–370, 2015.

[10] Y. Xie, *Dynamic Documents with R and knitr*, 2nd. Boca Raton, Florida: Chapman and Hall/CRC, 2015, ISBN 978-1498716963. [Online]. Available: `https://yihui.org/knitr/`.

[11] H. Wickham, *ggplot2: Elegant Graphics for Data Analysis*. Springer-Verlag New York, 2016, ISBN: 978-3-319-24277-4. [Online]. Available: `https://ggplot2.tidyverse.org`.

[12] C. Sievert, *Interactive Web-Based Data Visualization with R, plotly, and shiny*. Chapman and Hall/CRC, 2020, ISBN: 9781138331457. [Online]. Available: `https://plotly-r.com`.

[13] W. Chang et al., *Shiny: Web application framework for r*, R package version 1.11.1, 2025. DOI: `10.32614/CRAN.package.shiny`.

[14] D. Murdoch and D. Adler, *Rgl: 3d visualization using opengl*, R package version 1.3.24, 2025. DOI: `10.32614/CRAN.package.rgl`.